

Natural Language Processing of Maintenance Records Data

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ABSTRACT

Enterprise resource planning systems and maintenance management systems are commonly used by organisations for handling of maintenance records, through a graphical user interface. A maintenance record consists of a number of data fields, such as drop-down lists, list boxes, check boxes and text entry fields. In contrast to the other data fields, the operator has the freedom to type in any text in the text entry fields, to complement and make the maintenance record as complete as possible. Accordingly, the text entry fields of maintenance records can contain any words, in any number.

Data quality is crucial in statistical analysis of maintenance records, and therefore manual analysis of maintenance records' text entry fields is often necessary before any decision making. However, this may be a very tedious and resource consuming process.

In this article, natural language processing is applied to text entry fields of maintenance records in a case study, to show how it can bring further value in the assessment of technical assets' performance.

Keywords: Maintenance records, Natural language processing, Structured and unstructured data, Data quality, Rail infrastructure.

1. Introduction

Maintenance can be described as the combination of all technical and administrative actions, including supervision actions, intended to retain an item in, or restore it to, a state in which it can perform a required function [4, 6]. Maintenance can be divided into preventive and corrective maintenance, where both generally are followed up regarding performance and costs. Information technology (IT), such as enterprise resource planning (ERP) systems and maintenance management systems (MMS), are used for such activities. The data of preventive and corrective maintenance work is commonly called maintenance records, reports or work orders, and follows a set template and procedure for registration and closure, through a graphical user interface (GUI). Maintenance records contain a number of fields/boxes, such as: record identification number; asset information regarding system, subsystem and components; maintenance activity; failure cause; and remedy. However, the content depends if it is corrective or preventive maintenance records. The records fields within a GUI comprise of drop-down lists, list boxes, check boxes and text entry fields. In contrast to the other data fields, the text entry fields are filled as the operator thinks is necessary for the understanding of the work carried out. Accordingly, the text entry fields of maintenance records can contain any words, in any number, i.e. unstructured text.

High quality information depends on the quality of the raw data and the way it is processed [12]. For monitoring maintenance performance and costs, computerized analysis and eMaintenance solutions [9, 10] are applied by organizations, and especially in asset intensive or safety oriented organizations, e.g. manufacturing, transportation, aviation and nuclear. However, when it

comes to maintenance records, manual analysis of the records' text entry fields is normally required before any decision making. Specifically, it means reading records one by one, which is a tedious and resource consuming process. By computerized analysis, i.e. natural language processing (NLP), it is possible to make the process considerably more efficient. Also, data quality issues are to a great extent related to manual input and human errors. Aljumaili et al. [1] found that about 80% of data quality issues are related to human errors, while 20% are related to machine failures. Thus, NLP is also relevant in the sense that it has the potential to improve the data quality of such entry fields.

In this article, NLP is applied to maintenance records' text entry fields, as a case study. The aim is to demonstrate how basic NLP can bring further value in the assessment of technical assets' performance, by relating text entry field data to other data fields. After stressing the importance and introducing some of the main features of data quality, NLP and the method applied are described, followed by a case study on linear assets, or more specifically on railways. However, the method is generic and similar for other technical assets and organizations.

2. Data quality

Lack of relevant data and information is one of the main problems for decision-making within the maintenance process [10]. The provision of the right information to the right user with the right quality and at the right time is essential [7, 10]. High-quality data are commonly defined as data that are appropriate for use [11, 12]. Wang [12] presents a framework of data quality consisting of four categories: intrinsic, contextual, and representational and accessibility, see Table 1. For example, objectivity is the extent to which data are unbiased (un-prejudiced). An indica-

tor lacking objectivity could be one where a certain percentage of the data has been excluded without sound statistical reasoning. Excluded data could be long down times or down times of a specific system, which would make the indicator result appear good in certain perspectives.

Table 1. Data quality dimensions [12]

Category	Dimension
Intrinsic	Believability, Accuracy, Objectivity, Reputation
Contextual	Value-added, Relevancy, Timeliness, Completeness, Appropriate amount of data
Representational	Interpretability, Ease of understanding, Representational consistency, Concise representation
Accessibility	Accessibility, Access security

Quality dimensions with particular relevancy to text entry fields of maintenance records are: accessibility, representational consistency and completeness. Accessibility is the extent to which data are available or easily and quickly retrievable. Data that requires extensive manual intervention to be collected has poor accessibility, e.g. manual reading of text entry fields. Representational consistency is the extent to which data are presented in the same format and are compatible with previous data. Consequently, an input field which is free to fill out, i.e. text entry fields, results in such data quality issues. Completeness is the extent to which data are of sufficient breadth, depth and scope for the task at hand. Consequently, decision based on a set of indicators that do not consider maintenance records' text entry fields for description may not be as effective as desired. For example, in maintenance records, failures are commonly connected to systems and components using predefined drop-down lists. By sorting the records according to the systems and components, the items with the highest failure frequencies and down times can be found out. However, a component that is not predefined in a drop down list may go undetected.

3. Methodology

This section aims to cover the basics of NLP, together with the method applied to maintenance records. Possibilities offered by more advanced NLP methods are given in the discussion section.

3.1. Natural language processing (NLP)

NLP can be described as any kind of computer manipulation of natural language [3], i.e. it includes computer science and linguistics. As an example, a simple NLP can be to count word frequencies, while a highly advanced NLP could be to answer human-language questions. Examples of NLP applications are web search engines, machine translation and subject applications, such as medical records and enterprise data. A few applications within maintenance can be found. Bayoumi et al. [2] developed a NLP model for connecting maintenance faults data to vehicle sensor data. By use of NLP, it was found that maintenance faults can be described by a significantly reduced lexicon of words, and thereby improve retention rate of NLP to near 100%. Another NLP application is on automatic quality control of vehicle accident reports by Gerber and Tang [5]. The developed model identifies human introduced errors in accident reports, and showed to outperform the baseline methods used.

As a brief introduction to NLP, common terms and NLP processes are given below:

Corpus: A corpus is a large body of text; raw or categorized. A corpus can be used to for code training.

Sentence segmentation: Separation of sentences; difficult as a period is also used to mark abbreviations.

Tokenization: Dividing text into words, commas, etc., i.e. tokens.

Normalization of tokens: Change of upper case to lower case.

Stemming: Removal of affixes, e.g. -s and -ed.

Lemmaization: Mapping of various forms/inflections of words/tokens, e.g. good is the lemma of better. Lemmaization is also related to mapping of synonyms.

Part-of-speech tagging: Grouping into classes, e.g. nouns, verbs, adjectives and adverbs.

Chunking: Segmentation and labeling of tokens to entities, e.g. "She saw the black swan" consist of two noun phrase chunks, which are "she" and "the black swan".

An example of a corpus is the Brown Corpus, which was compiled in the 1960s out of about 500 text sources and contains a bit above one million terms, categorized and tagged. Another example is the Universal Declaration of Human Rights (UDHR) corpus, given in 372 languages. Even though many corpora are available, specific applications and languages often require compilation of a customized corpus. For introductory literature to NLP, see for example Manning and Schütze [8] and Bird et al. [3].

3.2. Method

Various computer programs for NLP are available with corpus, special modules and special applications, such as enterprise data. However, since the aim is to demonstrate how basic NLP can be applied to maintenance records, a customized algorithm has been written in MATLAB. A customized code can in contrast to special program commands demonstrate the process in more detail. Also, the application requires a customized corpus; maintenance data are of a particular asset in a particular language, i.e. three factors requiring specially made code.

For analysis of maintenance data text entry fields, the main steps of the algorithm, which is basic in programming, are shown in Figure 1.

The occurrence of token types can provide information about failure causes, types of failures, and the items that have more failures than others. Therefore, these extracted tokens, found to be interesting, can then be compared and linked to analysis of the other data fields, for additional information and study of agreement, or disagreement. Thus, the maintenance data is a semi-structured corpus.

4. Case study

A case study has been carried out on rail infrastructure to demonstrate the method discussed. The data used in this study was provided by Trafikverket (Swedish Transport Administration). However, analysis in other organizations and for other assets will not yield the same result, but the method for assessing is similar.

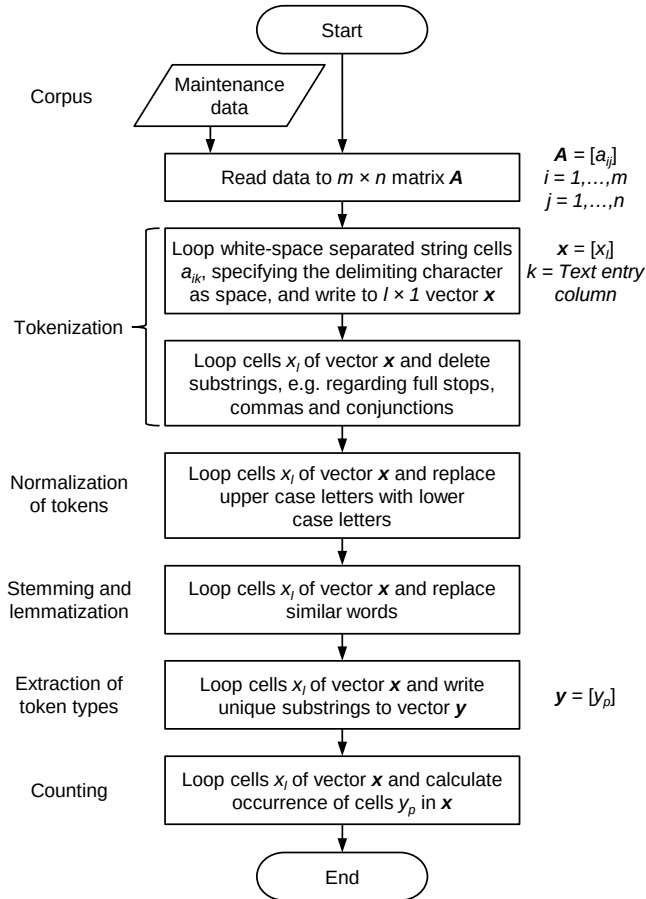


Figure 1: Flowchart of NLP algorithm.

4.1. Data collection

Operation and maintenance data have been collected from the Swedish railway section 111. Section 111 is a 128 km 30 tons axle load mixed traffic section of the Swedish Iron Ore Line, stretching from the border of Norway, Riksgränsen, to Kiruna city (Figure 2).

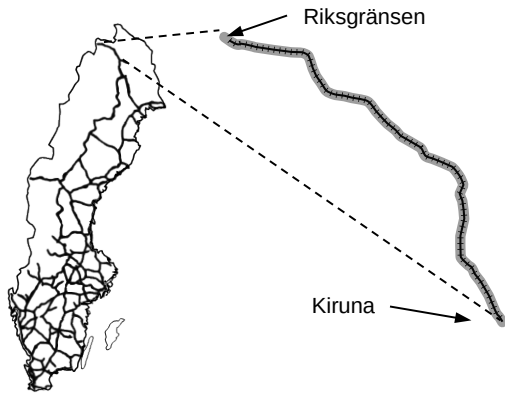


Figure 2: Swedish railway section 111 stretching from the border of Norway, Riksgränsen, to Kiruna city.

The failure data is collected from Trafikverket and constitute of infrastructure related corrective maintenance work, i.e. failure data. The corrective maintenance consist of urgent inspection remarks reported by the maintenance contractor, as well as failure events and failure symptoms identified outside the inspections, commonly reported by the train driver, but occasionally

reported by the public. The failure data is from 2001.01.01 – 2014.01.01, i.e. 13 years, which in total gives 10 958 records, with about one fourth causing train delays. The train delaying failures per system are shown in Figure 3.

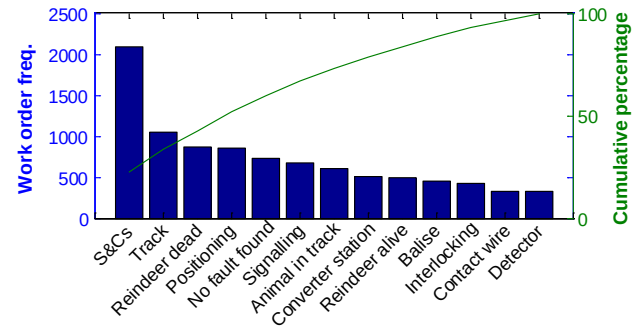


Figure 3: Failures per system of Section 111. S&Cs equals switches and crossings.

4.2. Data quality of the maintenance records

Simple data quality checks have been carried out on the failure data (Figure 4). Each record consists of 71 fields. Fields with 100 % usage means that all records have some text or numbers filled in. Therefore, a data field with low usage can mean that the data quality is low, or it may just not be applicable for every record. However, some data fields may be missed during the input process due to some error. As an example, the field for registering the failed component has a usage of ~30%. This, apparently low value, could be by several reasons, such as: failures cannot be allocated to a single component, like in the case of snow in switches and crossings (S&Cs); or that the component does not have a single commonly used name. However, the figure gives information regarding which data of the records that is suitable for case studies. Also, in this way it is possible to improve the work order process, e.g. by removing unnecessary fields and improving the way of completing other fields.

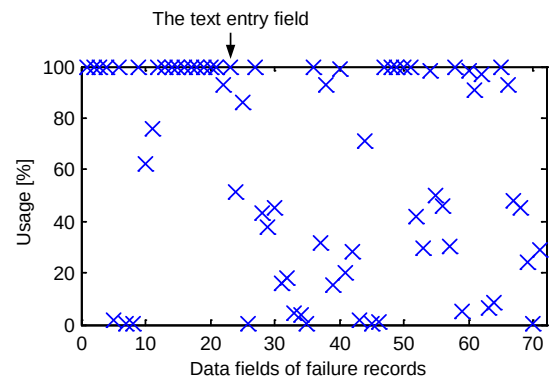


Figure 4: Usage of fields in the failure records.

The text entry field for describing the failure and work carried out is used in more than 99 % of the records (Figure 4). Thus, further analysis can be done as the field is frequently used.

4.3. Results and discussion

The text entry fields of the 10 958 records is found to contain 69 382 words in total. Following tokenization, normalization of tokens, stemming and extraction of token types (types), the number of types is found to be 8 442; see Table 2.

Table 2: Richness of words of the text entry field.

Total number of tokens in the text entry field, i.e. the description field	69 382
Number of token types, i.e. disregarding repetitions	11 400
Number of token types after removing commas and full stops	9 756
Number of token types after changing capital letters to lower case	8 442

By sorting the types by occurrence, it is noticed that the most frequently used types are found in more than one thousand of the records; see Figure 5. Furthermore, by limiting the study to the 250 most used types, it is seen that the 250th type occurred in 39 records, i.e. not many in comparison to the total number of records. After removing needless types, e.g. conjunctions and prepositions, 143 unique types are left of the 250. Finally, through grouping of similar words (stemming and lemmatization), e.g. singular, plural and synonyms, there are 104 words left. Figure 5 shows the first 80 types (unique “words”) with the highest occurrence. The terms are translated from Swedish to English, and consequently, the translated terms can consist of several words, like “error code” is one word in Swedish. A number of types are marked out by arrows in Figure 5 for discussion.

The second highest occurring type, “control”, means that switch points are not in control/position, and thus, the type “control” could be aggregated with the fifth most used type, “switch”, which then would become the most occurring type. By comparing Figures 5 and 3, it can be seen that the top token types and failures per system are similar, i.e. S&Cs and track.

The type “moose” is found on the 16th place, occurring in 234 records (cows, bulls and calves included as similar terms). By studying these 234 records, it is found that it would be hard to manually identify these records from the 10 958 records. The data can manually be sorted on animals in track and on the animal moose, but it would only give 149.

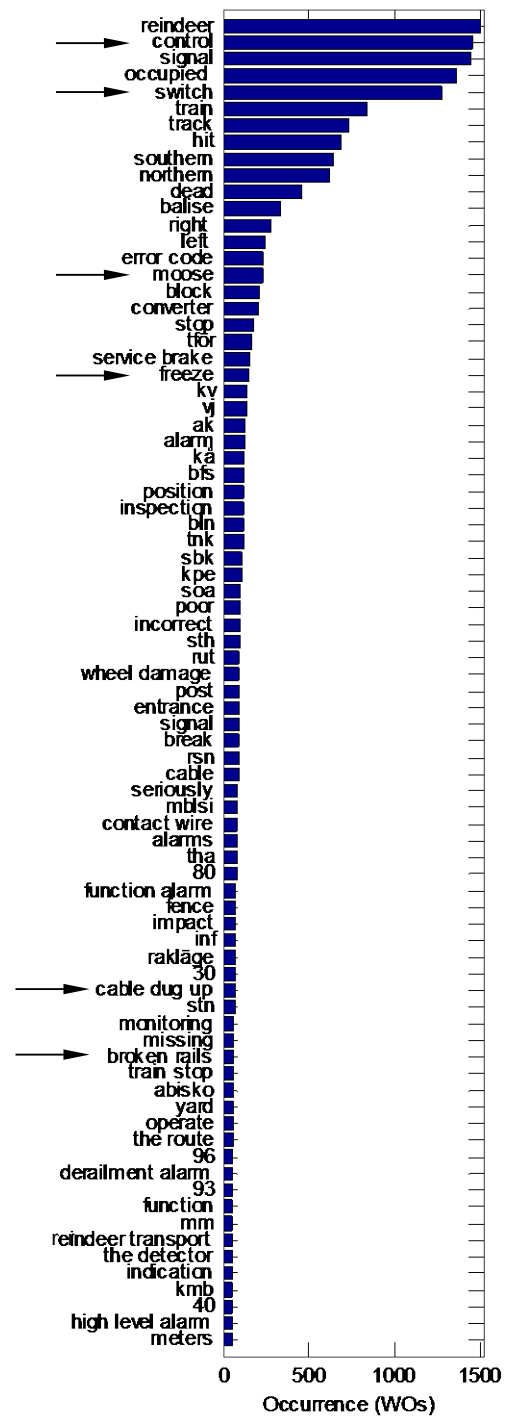
Another type is “freeze”, which occurred 144 times. Freeze is referring to computer freeze/hang. By studying the 144 records, it is found that it would not be possible to sort them out without reading the free entry field of each of the 10 958 records, i.e. 69 382 words.

The next type is “cable dug up” (often costly), occurring in 68 records. By manually sorting the 10 958 records for cable systems, 172 records would be found, which would include 47 of the 68 cable dug up. In other words, NLP gives additional information.

The type “Broken rails” was found in 63 records. By manually sorting for the predefined safety issue “Rail breakage”, 72 records are found. However, the NLP found 18 records that the manual sorting missed. Suspected rail breaks are excluded from these 18 records, but since the predefined safety issue “rail breakage” has not been ticked, there is still some uncertainty about some of these records, i.e. those not clearly described in the text entry fields.

The types “96” and “93” (not marked with arrows) are identification of trains, which make it possible to compare which trains are linked to most rail infrastructure failures.

Lastly, the type “derailment” (not shown in Figure 5) gives six records where wheels derailed. Since the MMS system did not have a specific box implemented for indicating derailment before 2009, only two of the records can easily be sorted out manually. Four records would be possible to identify by use of several predefined drop-down lists.

**Figure 5:** Occurrence of types (unique “words”) in the text entry field of maintenance records.

Results from the NLP give statistics on types (occurrence of unique “words”) in maintenance records’ text entry field, i.e. the descriptive text typed in by operators. Subsequently, the types have been used to extract maintenance records according to a specific type in the text entry field. The extracted records have then been compared by trying to extract the same records from the whole dataset manually, by use of filtering functions in spreadsheet software. However, spreadsheets are filtered by use of drop-down lists, or pivot tables, i.e. it cannot filter text entry fields. The comparison showed that some information can only be found in the text entry field, and consequently, manual reading of the text entry field is required to capture all records related to a specific failure type or unique word. For the data used in this

study, in a worst case scenario, it would mean reading text entry fields of 10 958 records, which equals 69 382 words. Alternatively, NLP can be carried out, which is fast and simple when algorithms are in place. It is clear from this study that NLP can save time in the analysis of data. In addition, more information can be extracted that can support decision making.

5. Conclusions

NLP of text entry fields of maintenance records has been demonstrated in a case study. It has been found that NLP makes the analysis process more efficient, gives additional information, and in some cases, it is the only realistic method for analysis of maintenance records, as long as resources for manual analysis is not endless. Moreover, since NLP improves identification of failures, it improves the input data to reliability and availability studies, as missing observations (failures) can have a large effect on the mean time between failures and maintenance times.

The method also provides an overview of data quality, as maintenance records extracted by applying NLP to the records' text entry fields gives information on what data is missing or is important in the records' predefined fields.

Case study specific results are as follows: NLP gave 68 cable dug up related maintenance records; manual extraction gave 47. NLP gave 144 computer freeze records; manual extraction was not possible. NLP gave six derailment records; manual extraction gave four. NLP gave 18 additional rail breaks to the 72 found manually. NLP gave 234 moose related records (rail infrastructure failures); manual extraction gave 149.

Acknowledgments

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